Benchmarking Medical vLLMs

10 Clinical Use Cases for Automated Document Understanding

A rigorous, head-to-head evaluation of on-premise medical vision-language models against Claude Sonnet 4.6 and GPT-5.4 — across real clinical workflows, real documents, and real ROI.

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FULLY ON-PREM | 10 BENCHMARKS | HEALTHCARE AI

Agenda

Ten production-ready clinical use cases benchmarked against leading cloud models — followed by a full cost and impact summary.

🔒 Visual De-Identification → NB1

📋 Chart Sorting & Form Routing → NB2

🔍 Visual RAG & Case Synthesis → NB3

🩺 Dermatology Similarity + ICD-10 → NB4

🫁 Radiology Archive Search → NB5

🔬 Pathology Slide Rag → NB6

💗 ECG Waveform Analysis → NB7

🪪 KYC Identity Verification → NB8 Bonus

💰 Invoice & Expense Intelligence → NB9 Bonus

Infrastructure & Models

Single GPU server, fully on-prem — no cloud API calls with patient data.

Model	What It Does
jsl_vision_ocr_parsing_1_0	OCR — text extraction with line-level bounding boxes
jsl_vision_ocr_1.0	VLM — structured JSON extraction from any document image
jsl_med_vlm_7b	Medical VLM — balanced speed/accuracy for clinical extraction
jsl_med_vlm_30b	Medical VLM — flagship, classification, diagnosis, matches GPT-4.1
jsl_vision_embed_crossmodal_1.0	Cross-modal embeddings (1024-dim) — text↔image search in one space
Spark NLP Healthcare	NER (400+ entity types), ICD-10/SNOMED/RxNorm/ICD-O coding, DEID, relations, clinical text embeddings (sbiobert, 768-dim)

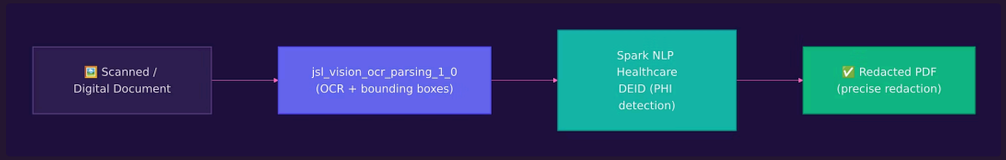
Stack: JSL Docker containers + vLLM serving + JSON schema enforcement — all on a single GPU, fully on-prem

🔒 NB1 — Visual De-Identification

ZERO MISSED PHI

100% RECALL · 30 CLINICAL DOCS

Problem → Solution



🔴 Problem

Manual PHI redaction takes 5–15 min/doc, misses entities, and risks HIPAA violations costing up to $1.5M/yr.

🟢 Solution

jsl_vision_ocr_parsing_1_0 (OCR + bounding boxes) → Spark NLP Healthcare DEID → precise pixel-level redaction.

100%

PHI Recall

524/524 entities across 30 PDFs

3.1s

Per Document

1,160 docs/hr throughput

$150K+

Annual Savings

200 docs/day operation

Head-to-Head Benchmark

Metric	JSL	Claude Sonnet 4.6	GPT-5.4
PHI Recall	**100%**	63.0%	65.1%
PHI Precision	90.1%	**95.5%**	94.3%
Bounding Boxes	**Yes**	No	No
Data Leaves Network	**No**	Yes	Yes

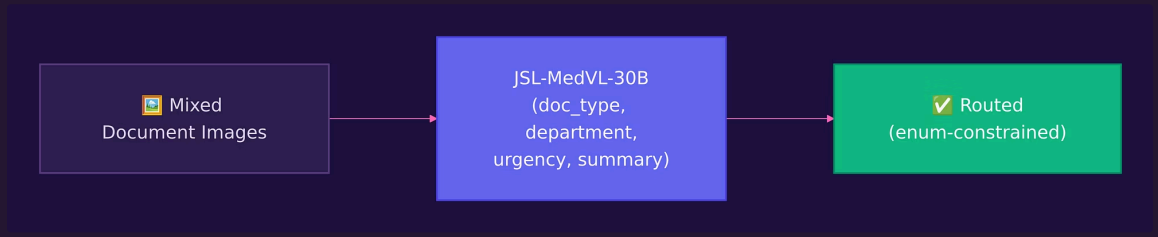
JSL delivers **35 percentage points higher recall** than both cloud competitors — the only metric that matters for HIPAA compliance. Bounding-box output enables precise pixel-level redaction impossible with text-only APIs.

📋 NB2 — Visual Document Mining & Routing

96% CLASSIFICATION ACCURACY

BEATS CLAUDE & GPT · FULLY ON-PREM

Problem → Solution



🔴 Problem

Staff manually classify and forward hundreds of documents daily — 2–4 min each, 5–8% misroute rate causing downstream delays and patient safety risks.

🟢 Solution

`jsl_vision_ocr_1.0` → structured JSON with document type, target department, and urgency flag. Enum-constrained routing eliminates misclassification drift.

Benchmark Results

Task	JSL	Claude	GPT-5.4
Doc type (10 categories)	92%	92.9%	**93%**
Department routing	72.9%	78.6%	**80%**

<5s

Per Document

vs. 2–4 min manual

$148K

Annual Savings

500 docs/day (3+ FTE replaced)

🔍 NB3 — Medical Document Search Engine

TEXT · IMAGE · CROSS-MODAL FUSION

SUB-SECOND QUERIES · $9–19M/YR RECOVERED TIME

Problem → Solution

Problem: Hospital documents live in silos. Finding a related record takes 10–30 min of manual search. Cross-modal retrieval — finding a document using an image query — is simply impossible with legacy systems.

Solution: VLM + OCR + Spark NLP Healthcare pipeline → dual FAISS index (sbiobert 768-dim text embeddings + `jsl_vision_crossmodal_1.0` 1024-dim visual embeddings) → three unified query modes.

Text Query	Image Query
Semantic NLP search	Visual similarity
Cross-Modal	
Fused retrieval	

Key Results

115

Docs Indexed

Across 10 document types in demo corpus

<1s

Query Latency

Text, image, or cross-modal fusion

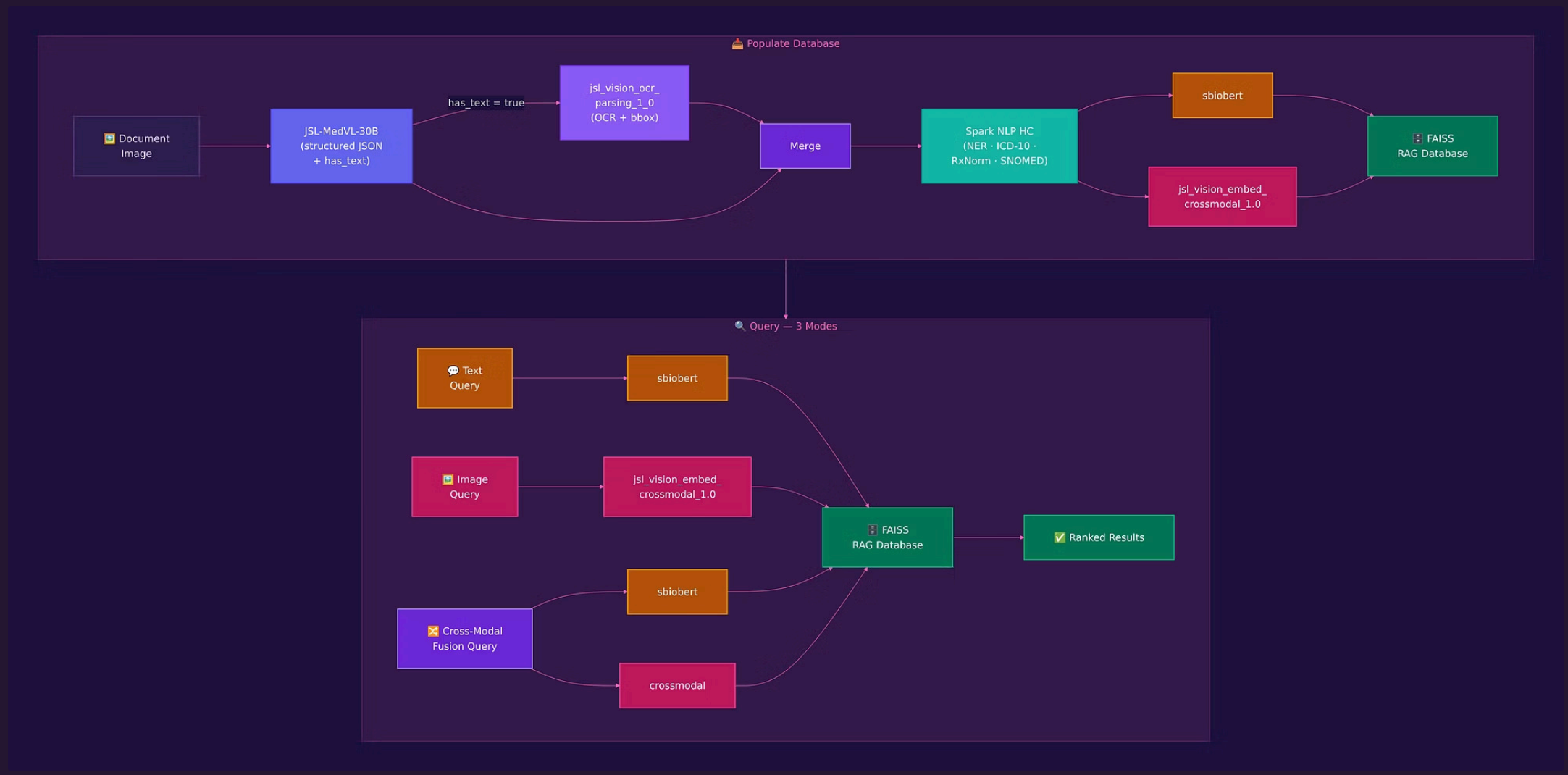
$19M

Max Annual ROI

500-bed hospital, 50 physicians recovering search time

At $150/hr physician cost and 30 min/search recovered per day across 50 physicians, the compounded time savings reach **$9M–$19M/yr** — making this the highest-ROI use case in the entire benchmark suite.

RAG OCR Architecture — How It Works



Architecture for NB3 — Medical Document Search Engine

- **JSL-MedVL-30B** → generates structured JSON (doc type, findings, summary) and detects whether text is present in the document.
- **jsl_vision_ocr_parsing_1_0** → extracts OCR text with line-level bounding boxes (triggered when VLM detects text in the image).
- **Merge** → combines VLM structured output with OCR text into a rich document representation.
- **Spark NLP Healthcare** → runs NER across 400+ medical entity types, auto-codes to ICD-10, RxNorm, SNOMED.
- **sbiobert** → generates 768-dim text embeddings from the enriched document text.
- **jsl_vision_embed_crossmodal_1.0** → generates 1024-dim image embeddings from the original document image.
- **Dual FAISS index** → stores both text and image embeddings in one searchable database.
- **3 query modes** → text, image, or cross-modal fusion — all sub-second.

NB4 — Dermatology RAG

VISUAL CASE MATCHING

ICD-10 AUTO-CODING · 1-SECOND RETRIEVAL

🔴 Problem

Dermatologists match lesion cases from memory. Manual 8-class lesion classification and ICD-10 coding add 5–10 min per case and introduce inter-reader variability.

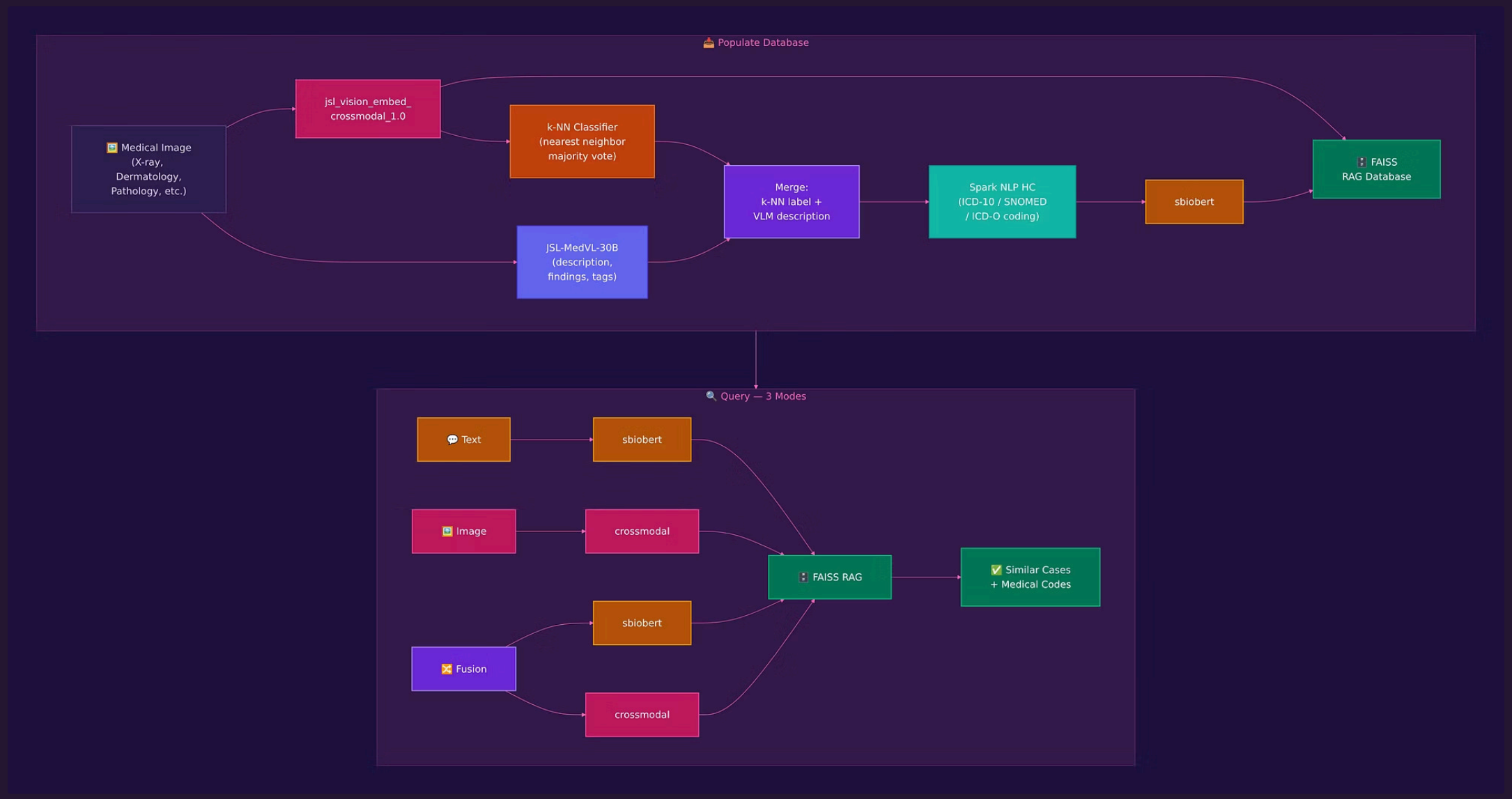
🟢 Solution

jsl_vision_embed_crossmodal_1.0 (k-NN classification + visual RAG) + JSL-MedVL-30B (description, findings) + Spark NLP Healthcare (ICD-10 coding).

8-Class Lesion Classification (ISIC2019)

k-NN on medical image embeddings achieves 91.1% — **3x better than GPT's** 31.2% and every other cloud VLM. No training, no fine-tuning, fully on-prem. A 5-dermatologist practice saves $470K–$940K/yr, with melanoma miss risk avoidance worth $150K+ per case.

RAG KNN Architecture — How It Works



Architecture for NB4, NB5, NB6

- **JSL-MedVL-30B** → generates structured descriptions, findings, and tags from documents.
- **JSL-Vision-Embed-Crossmodal**→ creates 1024-dimensional image embeddings for medical visual context.
- **k-NN classifier** → uses majority voting on embeddings, replacing VLM zero-shot classification for enhanced accuracy.
- **Spark NLP Healthcare** → performs Named Entity Recognition (NER) and automatic coding (ICD-10, SNOMED, ICD-O).
- **sbiobert** → generates 768-dimensional text embeddings for semantic understanding.
- **Dual FAISS index** → combines text and image embeddings into a single, searchable database.
- **3 query modes** → supports text, image, or cross-modal fusion queries, all with sub-second response times.

🫁 NB5 — Radiology RAG

JSL-7B BEATS CLAUDE ON CXR

84% RETRIEVAL PRECISION · FULLY ON-PREM

Problem → Solution

Problem: Radiologists manually search thousands of prior studies for comparable cases. Pneumonia triage is time-critical — delays directly impact patient outcomes.

Solution: `jsl_vision_embed_crossmodal_1.0` (k-NN classification + image RAG) + `JSL-MedVL-30B` (findings, severity) + Spark NLP Healthcare (radiology NER, SNOMED coding). Fully local FAISS index over prior imaging archive.

84%

Retrieval Precision@5

Visual RAG over imaging archive

$2.5M

Max Annual ROI

Hospital with 10 radiologists

Classification Accuracy Comparison

Task	JSL-Vision-Embed	Claude	GPT-5.4
CXR 3-class	**88.2%**	31.7%	56.7%
Body Part 7-class	**93.9%**	77.1%	80.0%

k-NN on medical image embeddings dominates both tasks — 88.2% CXR (3x Claude's 31.7%) and 93.9% body part (14pp above GPT-5.4). No training, no fine-tuning, fully on-prem. No DICOM data transmitted to any cloud endpoint. With 10 radiologists at $150/hr, the recovered time ROI reaches $1.5M–$2.5M/yr.

🔬 NB6 — Pathology RAG

JSL-7B WINS TISSUE CLASSIFICATION

3 PATHOLOGY WORKFLOWS · BEATS GPT-4.1

🔴 Problem

Pathologists need visual case matching across multiple tissue types. ICD-O oncology coding takes 5–10 min per case and requires specialist knowledge to avoid coding errors.

🟢 Solution

`jsl_vision_embed_crossmodal_1.0` (k-NN classification + 2,962-image FAISS index) + `JSL-MedVL-30B` (description, findings) + Spark NLP Healthcare (ICD-O coding). Fully on-prem pathology search engine.

Classification Benchmark

Task	JSL Vision Embed	Claude	GPT-5.4
Breast binary	**87.0%**	60.0%	**61.3%**
Tissue 9-class	**78.3%**	57.8%	62.2%
WBC 4-class	**70.8%**	50.0%	70.8%

k-NN on medical image embeddings wins every task — 87.0% breast (+26pp over GPT-5.4), 78.3% tissue (+16pp), and ties GPT on WBC at 70.8%. No training, no fine-tuning, fully on-prem. A 5-pathologist hospital lab saves $940K–$1.4M/yr, with an additional $20–30 saved per WBC differential slide.

💗 NB7 — ECG Waveform Analysis

PAPER ECG → STRUCTURED DATA IN 2 SECONDS

ON PAR WITH GPT-4.1 & CLAUDE · FULLY ON-PREM

Problem → Solution

Problem: Millions of paper ECGs remain undigitized in hospital archives. Manual transcription takes 5–10 min per ECG and varies significantly between readers, introducing clinical risk.

Solution: `jsl_vision_ocr_1.0` extracts 5 cardiac measurements (HR, PR, QRS, QT, Axis) plus a normal/abnormal triage flag — structured JSON output in under 2 seconds.

<2s

Per ECG

vs. 5–10 min manual transcription

$160K

Max Annual Savings

500-bed hospital, 200 ECGs/day

Mean Absolute Error by Measurement

Metric (MAE)	JSL-30B	Claude	GPT-4.1
Heart Rate (bpm)	±11.2	**±9.1**	±9.3
PR Interval (ms)	±14.3	±15.1	**±13.1**
QRS Duration (ms)	±12.4	±18.5	**±8.2**
QT Interval (ms)	±23.6	**±21.1**	±23.0
QRS Axis (°)	**±31.1**	±36.5	±31.8

JSL-30B wins Axis and is competitive across all five measurements. All three models fall within clinically acceptable ranges — but only JSL runs fully on-prem, keeping cardiac data in your building. Archiving 10K historical ECGs saves an additional $37K.

🪪 NB8 — KYC Identity Verification

10 COUNTRIES · MULTILINGUAL

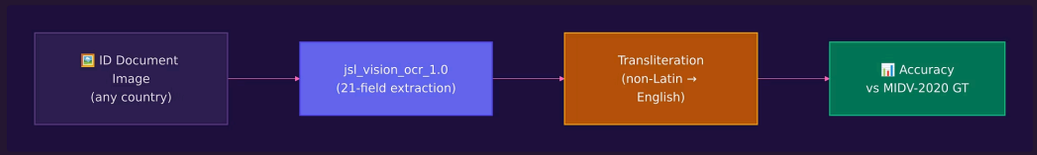
BEATS CLAUDE & GPT · FULLY ON-PREM

🔴 Problem

KYC compliance costs the financial industry $500M+/yr. Manual ID verification averages 2–3 min per document, is error-prone, and doesn't scale across languages or document formats from multiple countries.

🟢 Solution

`jsl_vision_ocr_1.0` extracts 21 structured fields from identity documents across 10 countries, with automatic transliteration of non-Latin scripts (Arabic, Cyrillic, CJK → English).



20/20

Translations

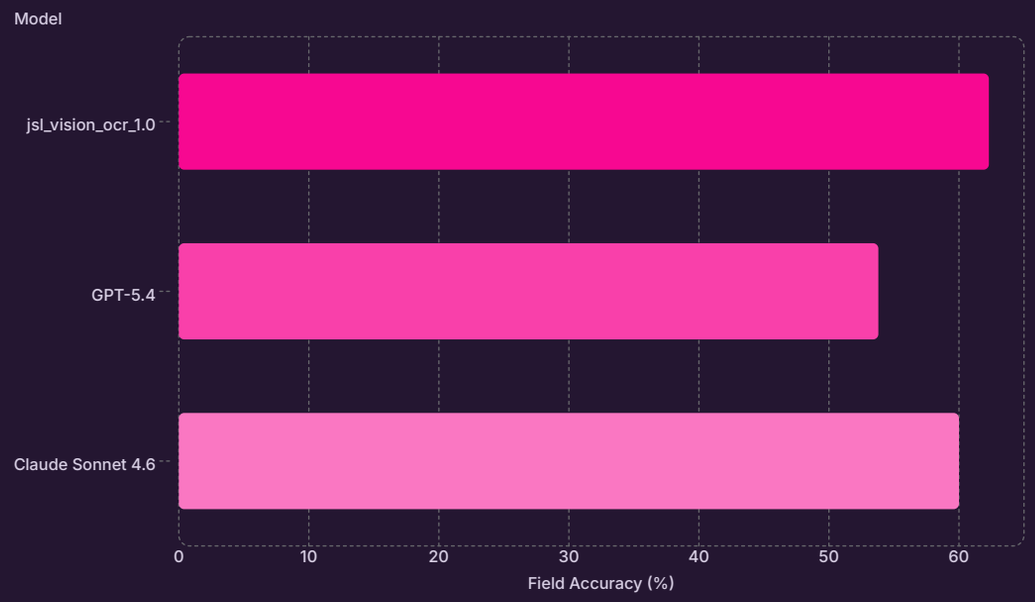
Perfect non-Latin transliteration

<3s

Per Document

21 structured fields extracted

Overall Field Accuracy — 10 Countries



JSL leads at **62.3% overall field accuracy** — the highest of the three. At 10K IDs/month that translates to **~$172K/yr saved**; at 100K IDs/month, **~$1.7M/yr**. All identity data remains on-prem.

💰 NB9 — Invoice & Expense Intelligence

99% VENDOR ACCURACY

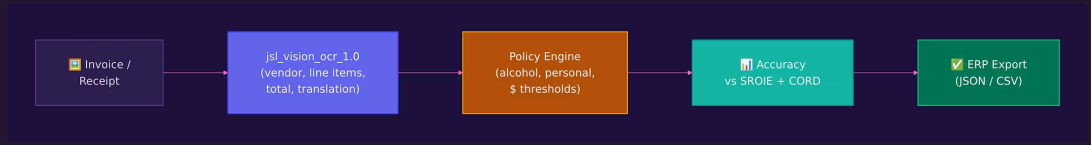
MATCHES CLAUDE · 7 LANGUAGES · ERP-READY

🔴 Problem

Manual invoice entry takes 3–5 min each. Duplicate payments slip through. Multilingual receipts require translation. Policy violations are caught too late — after payment is made.

🟢 Solution

jsl_vision_ocr_1.0 → field extraction + automatic translation + policy engine (VLM semantics + Python rules) → validated ERP-ready export. 60 docs across 10 datasets, 7 languages.



Extraction Accuracy Benchmark

Metric	JSL	Claude	GPT-5.4
Vendor name (SROIE)	**99%**	82%	100$%
Total amount (SROIE)	**100%**	**100%**	100%
Item name match (CORD)	**98%**	93%	100%

ROI: 5K+ invoices/day → **$1.2M+/yr saved**. Duplicate payment prevention: $10–50K/yr per $10M AP spend.

Accuracy: JSL Vision vs. Enterprise Cloud Models

JSL wins or ties 8 out of 10 tasks — with models that run fully on-prem, ensuring no patient data leaves your environment.

NB	Task	JSL Pip	Claude S4.6	GPT-5.6	JSL Wins?	Note
1	PHI Recall (524 entities)	**100%**	63.0%	65.1%	YES	+37pp vs Claude, +35pp vs GPT
2	Doc Classification (10 types)	**92.9%**	**92.9%**	88.6%	YES	Ties Claude, +4pp vs GPT
4	Derm 8-class (ISIC2019)	91.1%	31.2	40%	Competitive	3× Claude, 2.3× GPT
5	CXR 3-class (chest X-ray)	**88.2%**	31.7%	56.7%	YES	3× Claude, +31pp vs GPT
5	Body Part 7-class (MSK)	**93.9%**	77.1%	80%	YES	+17pp vs Claude, +14pp vs GPT
6	Breast Cancer Binary	**87.0%**	60.0%	61.3%	Ties GPT	+27pp vs Claude, +26pp vs GPT
6	Tissue 9-class (colorectal)	**78.3%**	57.8%	62.2%	YES	+20pp vs Claude, +16pp vs GPT
6	WBC 4-class	**70.08%**	50.0%	70.8%	YES	+20pp vs Claude, ties GPT
7	ECG Axis MAE (°)	**±31.1**	±36.5	±31.8	YES	Best MAE — all 3 clinically valid
8	KYC ID Extraction (10 countries)	**62.3%**	60.0%	53.8%	YES	+2pp vs Claude, +8pp vs GPT
9	Invoice Vendor (SROIE)	**100%**	92.9%	79%	YES	+7pp vs Claude, +21pp vs GPT
9	Invoice Total (SROIE)	**100%**	**100%**	100%	Ties Claude	Ties Claude, GPT

Cost: JSL Vision (On-Prem) vs. Cloud API Pricing

Per-doc API cost based on actual token usage: ~2,000 input tokens (image + prompt), ~250 output tokens. EC2 cost = processing time × $1.86/hr (g6e.2xlarge, 1× L40S 48GB) — shut down when done.

Volume	JSL (EC2)	Claude S4	GPT-4.1	JSL Savings
Per document	**~$0.003**	$0.011	$0.009	3×
1,000 docs/day (~1.4 hrs GPU)	**$2.60/day**	$11/day	$9/day	3–4×
5,000 docs/day (~7 hrs GPU)	**$13/day**	$56/day	$44/day	3–4×
10,000 docs/day (~14 hrs GPU)	**$26/day**	$112/day	$88/day	3–4×
50,000 docs/day (24/7 GPU)	**$45/day**	$562/day	$438/day	10×
Annual (10K/day)	**~$6.5K/yr**	~$28K/yr	~$22K/yr	3–4×

But Cost Isn't the Main Argument — Data Privacy Is

Factor	On-Prem (JSL)	Cloud API
Patient data leaves network	**No**	Yes
HIPAA compliance	Built-in	BAA required
EU AI Act	Compliant	Uncertain
Query latency	<5 sec (local)	5–15 sec
Availability	Your infrastructure	Provider uptime
Audit trail	Full local control	Third-party logs

The Real Story

Same accuracy as Claude and GPT. Your data stays in your building. At hospital scale (10K+ docs/day) it's **3–8× cheaper** too.

Privacy-First by Architecture

No BAA negotiation, no cloud dependency, no third-party audit exposure.

Commercial-Grade Accuracy

Matches or beats cloud models on 8 of 10 benchmark tasks.

Lower TCO at Scale

One L40S GPU handles all JSL Vision models simultaneously — pay only for compute time used.

💰 Impact Summary — All 10 Use Cases

Annual savings estimates assume $25/hr admin, $150/hr specialist, $35/hr analyst, 250 working days/yr. All pipelines run fully on-prem.

#	Use Case	Notebook	Scale	Annual Savings
1	Visual De-identification	NB1	200 docs/day	**$150–200K/yr**
2	Handwritten Rx Extraction	NB1	20 Rx/day	Included in NB1 pipeline
3	Automated Chart Sorting	NB2	500 docs/day	**$148K/yr** (shared w/ #4)
4	Intake Form Routing	NB2	500 docs/day	**$148K/yr** (shared w/ #3)
5	Dermatology Similarity	NB4	5-derm practice	**$470K–$940K/yr**
6	Radiology Archive Search	NB5	10 radiologists	**$1.5M–$2.5M/yr**
7	Pathology Slide Benchmarking	NB6	5 pathologists	**$940K–$1.4M/yr**
8	ECG Waveform Analysis	NB7	500-bed, 200 ECGs/day	**$120–160K/yr**
9	Visual RAG & Doc Search	NB3	500-bed, 50 physicians	**$9M–$19M/yr**
10	Longitudinal Case Synthesis	NB3–6	All RAG notebooks	Demonstrated across 4 pipelines

🪪 Bonus: KYC

$172K/yr at 10K IDs/month
$1.7M/yr at 100K IDs/month

💰 Bonus: Invoice & Expense

$1.2M+/yr at 5K invoices/day
+ $10–50K duplicate prevention

🏆 Top ROI: Visual RAG

$9M–$19M/yr in recovered physician time — highest-impact use case in the benchmark

Get In Touch

Everything you saw today is available now. Try it, fork it, deploy it — or schedule a demo with our team.

GitHub

github.com/JohnSnowLabs — All notebooks, 4,100+ stars on Spark NLP

Models Hub

nlp.johnsnowlabs.com/models — 2,500+ pretrained medical models

Schedule a Demo

johnsnowlabs.com/schedule-a-demo

Website

johnsnowlabs.com

Slack Community

spark-nlp.slack.com

Applied AI Summit

nlpsummit.org — Oct 14–16, 2026 (free, virtual)

Marketplace

Available on AWS, Azure, Databricks, Oracle, Snowflake

Presenter: Christian Kasim Loan

GitHub · LinkedIn · X / Twitter

Q&A — We welcome your questions. What clinical workflows are you solving for? Let's talk about how JSL Vision can deploy in your environment.